

Ancient Tamil Character Recognition from Stone Inscriptions – A Theoretical Analysis

S. Dhivya

Department of Computing Technologies
College of Engineering and Technology, Faculty of Engineering
and Technology,
SRM Institute of Science and Technology,
SRM Nagar, Kattankulathur, 603203
Kanchipuram, Chennai, TN, India
dhivs37@gmail.com

J. Rene Beulah

Department of Computing Technologies
College of Engineering and Technology, Faculty of Engineering
and Technology,
SRM Institute of Science and Technology,
SRM Nagar, Kattankulathur, 603203
Kanchipuram, Chennai, TN, India
renebeulah@gmail.com

Abstract- Character recognition on inscriptions is an area which explores our knowledge of an ancient language. Inscriptions were done in all kinds of environments. This work focuses on recognizing Tamil language characters on stone-based images. From the inscription images, we come to know about the importance of old century languages. Some of the general challenges researchers face in recognizing the characters in stone inscriptions are differentiating the foreground pixel from the background stone images, perspective distortion, different light illumination, the same kind of background/foreground, damaged stones, lack of shape and size of the text. Despite the different ways proposed by the researchers, obstacles and issues continue to exist. This survey attempts to give a detailed analysis of the recent research works on character recognition from the images of stone inscriptions with a special reference to Tamil. It details the methods applied for preprocessing, feature extraction and classification. It gives a road map for future researchers who wish to carry out research in this area.

Keywords —image processing, preprocessing, feature extraction, classification, optical character recognition, convolutional neural network.

I. INTRODUCTION

A. Significance of Stone Inscriptions

Inscriptions were engraved on stones, rocks, copper plates, seals, temple walls, coins, palm leaves and bricks. According to the Archaeology department, out of the 25,000 stone inscriptions which were found in Tamilnadu, 15,000 inscriptions are still to be recognized. Generally, when a new stone inscription is found, archaeologists will perform a manual method to take imprints of the inscriptions on papers by applying ink on top of the stone, which are called Estampages. This is the first step in deciphering the text on the inscriptions. Estampages are not properly preserved and maintained for a long time. It gets destroyed and the valuable information in it is lost. Nowadays, stone inscriptions are easily captured as digital images. Stone inscriptions are sources of information about the life style of people of different ages. They give details like socio-economic conditions, cultural status, literature, art, life system, administration, astrology, medicine and education. Automatic recognition and translation will help today's younger generation who are not aware of this

information to easily understand the historical information and preserve it in digitized format [1].

B. Evolution of Tamil Script

Tamil is one of the world's oldest languages having evolved over centuries. There are alphabet systems such as sangakala Tamil, karoshi, Brahmi, vatteluttu, grindham and modern character set [2-3]. The Brahmi script is developed into developing northern scripts called Oriya, Devanagiri, Gurmukhi, Bengali, Gujarati and southern scripts called Telugu, Malayalam, Tamil, Kannada. The vattezhuthu started from 5th to 7th century. From the 8th century it developed into modern Tamil characters. Finally, modern Tamil scripts have 247 characters. Tamil characters deal with context complications, mismatch of locations, variation in directions, difference in shape and discontinuities in structures. Of the 247 characters, there are 12 vowels, 1 aayudham, 18 consonants and 216 characters which are a combination of consonants and vowels.

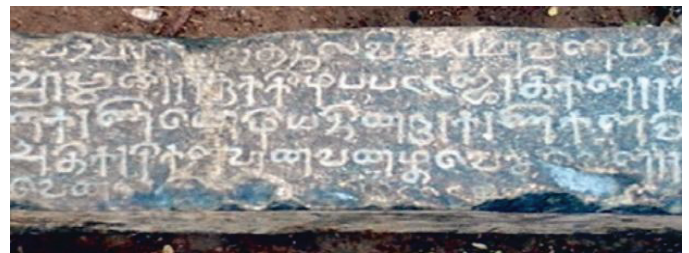


Fig. 1. Chola stone inscription found near Thanjavur

Fig.1 shows an ancient Chola stone inscription found near Thanjavur and Fig.2 shows the development of ancient Tamil characters. The present generation cannot identify the olden Tamil letters found in most of such inscriptions. Many research works are focused on modernizing, translating and dating the inscriptions so that the message conveyed by the ancient Tamil Scholars and Physicians can reach people of all generations.

History of Tamil Script									
சாதிக்கம்	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
Century	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
BC 3 rd C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 2 nd C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 3 rd C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 4 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 5 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 6 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 7 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 8 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 9 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 10 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 11 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 12 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 13 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 14 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 15 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 16 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 17 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 18 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ
AD 19 th C	அ	ஆ	இ	ஊ	ஐ	ஓ	ஔ	ஶ	ஸ

Fig. 2. Development of Ancient Tamil characters

C. Optical Character Recognition

The method of recognizing and digitizing text is known as Optical Character Recognition[4]. It helps to reduce our human effort and mistakes which are made manually. In the early days when OCR was introduced, it can work with only specific fonts. Recently the OCR methods are more sophisticated that they can even recognize people's handwriting from documents, which is termed as intelligent character recognition. OCR systems are applicable for license plate recognition, smart card processing, automatic data entry, DD processing, address/zip code recognition, handwritten character identification etc. Efficient extraction and storing information from documents, receipts, invoices, passports etc. OCR also helps in recognizing characters in a scanned document like book thereby converting raw images into editable text.

OCR can help automate operations in a number of ways like converting read-only files to text that can be edited, creating files that can be heard, translating from one language to another, and managing forms and questionnaires[5]. In addition, OCR assists in increasing the speed and accuracy of data entry where a digital pen is used and the stroke details and pen location are assessed while moving around the environment.

The OCR methods have a drawback that they are not appropriate for blurred images and images with noisy backgrounds and they will not yield good recognition rate. This paved way for using machine learning and deep learning methods for Character Recognition.

D. Machine Learning & Deep Learning for Character Recognition

Machine Learning and Deep Learning techniques are being applied for character recognition for more than a decade to achieve robustness, that is, to recognize characters from noisy and blurred backgrounds[6].

Machine learning algorithms get trained from the input data in a short time because of less number of layers involved, while deep learning approaches involve more number of layers, which make the learning more accurate[5]. Also, in machine learning, feature extraction is handled separately while in deep

learning, features are extracted automatically during the learning phase.

Because of the promising features of deep learning has become the most commonly used method for automatic character recognition since its inception[7]. Deep learning has unequalled power in solving complex problems and is clearly the right choice for automatic character recognition.

This paper provides a detailed review of research studies that have employed machine learning / deep learning for recognizing characters from stone inscriptions with a special emphasis on Tamil inscriptions. Also, the general framework of such methods is discussed in detail with the steps involved.

Section II lists the related research works. Section III details the steps involved in automatic character recognition with a general framework. Section IV gives a detailed discussion and analysis of the existing methods which will guide future researchers in this field to concentrate on the areas which need improvement. Section V gives the concluding remarks.

II. RELATED WORK

Shanmugametal. have applied deep convolution neural network with extreme learning based Newton metaheuristic method for extracting features and for recognizing characters. Input images are preprocessed through binarization and filtering methods. Tsallis entropy-based atom search algorithm is used to identify boundaries with lines, curves, bends etc.[7]. The method is applied on Tamil Handwritten character dataset by HP lab. In every class, 500 samples were taken with different styles of characters.

Preethiet al. have proposed an automated way of segmenting and extracting characters on inscriptions by applying nearest neighbor clustering and SVM classifier [8]. Linear discriminant analysis is used for reducing the dimensionality. The authors collected dataset from the Archaeological Survey of India, Mysore and it covers inscriptions from the 4th century to 15th century.

Manigandan et al. have applied SVM classifier on the 9th and 12th century inscription images collected from the Archaeological Department of Tamilnadu [2]. OCR and Natural Language Processing (NLP) techniques were used to map the identified character with a Unicode character set using pattern matching algorithm.

Thippeswamy et al. have applied decision tree algorithms for digitizing estampages of handwritten kannada stone inscription images [9]. Retinex and morphological operations are used to remove the noise, segmentation is performed by connected component labeling, local binary features are then extracted and finally decision tree is applied for classification. The method is tested on 11th century kannada stone inscription images.

Bhat et al. have proposed gaussian model-based background elimination using expectation maximization algorithm for enhancement and binarization of ancient scripts [10]. The method is tested on inscriptions gathered from

various parts of South India during the period from 4th century to 12th century.

Devi et al. have proposed an Image-based Character Pattern Identification (ICPI) system and Modified Speeded up Robust Feature with Bag of Grapheme (MSURF-BoG) algorithm is used for Feature Extraction[3]. The method was applied on 11th century images which consists of 17 consonants and 7 vowels.

Sathik et al. have proposed a Text line slicing method for palm leaf manuscriptsto segment touching text lines without modifying the original shape of the character while preserving the strokes [11].

Agarwal et al. have presented their study of machine learning techniques for identifying handwritten characters in Devanagari script [5].

Lincy et al. have applied Hierarchical SVM for recognizing Tamil characters from thousands of images of Tamil handwritings [12].

Robert Singh et al. have used Binary Robust Invariant Scalable Key Point algorithm to predict the strokes of Chinese characters. The method constructs octaves by sampling and rotation estimation [13].

Vinotheni et al. have applied modified convolutional neural network to achieve improvedcomputational efficiency and classification accuracy for offline mode character recognition, where characters are collected from various schools and tested [14].

Kaliappanet al. have suggested a hybrid classification methodwhich combines weighted contribution of feature-based approach and CNN to improve the classification and recognition of vowels in Tamil language [15].

BapuChandage et al. have presented their study on various algorithms used for preprocessing, segmentation, classification methods especially SVM, KNN, ANN, PNN classifiers for character recognition [16].

Ezhilarasi et al. have built a neural network for recognizing and Parts of Speech (POS) tagging on the 11th century stone inscription [17]. BidirectionalLong-Short Term Memory (Bi-LSTM) model is applied for classifying the characters.

Sinhala inscription characters are processed and classified by applying CNN and ANN with OCR models by Dias et al. [18]

Rajithkumar et al. have proposed Mean, Standard Deviation and Sum of Absolute Difference algorithm for Hoysala and Ganga period characters and have translated them into Kannada script[19].

Athisayamani et al. have applied Recurrent Neural Network (RNN) and Bi-LSTM network are used to identify the characters extracted from the palm leaf images [20].

A novel model of CNN is proposed by Ulaganathan et al for the classification of handwritten Tamil characters on printed documents [21]. A dataset consisting of 82928 sample images is used for testing.

Chendage et al. have written a study paper about Marathi scripts for efficient recognition. They claim that Principal Component Analysis and Radon transform for preprocessing leads to improve the efficiency of recognition rate [22].

Prabavathi et al. have focused on Tamil characters, both vowels and consonants [1]. During preprocessing, binarization technique is applied for separating the two colors, denoising is used to remove noise, line, word and character segmentation are used for getting clear features andBilinear interpolation normalization algorithm is applied to persist into specific size. Backpropagation deep neural network is used for classification.

Ranjanet et al. have focused on edge detection and proposed a fuzzy rule-based system that uses a Gabor filter to regulate the edge quality and Gaussian filter to reduce noise caused by sharpening [23].

Kauret al. have concentrated on Brahmi characters. Histogram of oriented gradient is used to extract features and Support Vector Machine is used for recognition of characters which helped in achieving good accuracy rate [24].

Chandrakala et al. have focused an automatic segmentation method for proper division of particular character [25]. Total variation regularization, an enhancement method is used to decompose the noise without blurring the actual sharp edges. The methods were applied on Kannada stone inscription image dataset which is available at the Archeological survey of India.

On reviewing the literature, it is inferred that OCR techniques were first used to recognize Tamil characters instone inscriptions. Nowadays, deep learning techniques are more prevalent in the research of character recognition. CNN is the most used model for classification. CNN architecture using ReLU activation function has yielded good segmentation rate and recognition rate by mapping the ancient characters to modern characters. Also, it is analyzed that preprocessing is an important role in improving the accuracy of character recognition and so many researchers have focused on only preprocessing techniques. A combo of the right methods for preprocessing and the right deep learning network will give much more accuracy.

III. FRAMEWORK FOR AUTOMATED CHARACTER RECOGNITION

When applying deep learning techniques for character recognition, the steps involved are Preprocessing,Segmentation,Feature extraction,Classification and Recognition as shown in Fig. 3.

A. Preprocessing

Preprocessing involves improving the clarity of images by applying image sharpening and restoration methods[10]. It enhances some features which are more relevant for further analysis. Outlier detection, handling duplicate and missing values and noise removal are also some other preprocessing steps which help the learning process to be unbiased and more accurate[26].Some of the preprocessing techniques are

1. Pixel brightness transformations/ corrections
2. Geometrical Transformations
3. Image Filtering and Segmentation

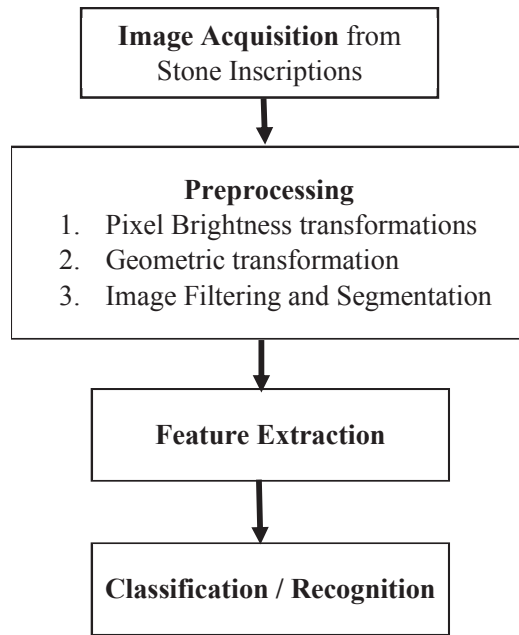


Fig. 3. Steps in Character Recognition

1) Pixel Brightness Transformations

The brightness transformations try to modify pixel brightness and the transformation depends on the properties of a pixel. The two types of Brightness transformations are brightness corrections and gray scale transformation[27].

$$O(x, y) = O1 + (O2 - O1)/(I2 - I1)[I(x, y) - I1]$$

where O1 tends to 0, O2 represents the number of levels (L-1=255), I1, I2 represents the min and max values of the grey level. Basically, adjusting brightness by adding bias value 'b', b>0 increases the brightness and b<0 darken the image. In addition, gain factor 'a' is added to the input pixel where a<1 darker image and a>1 brighter image.

$$O(x, y) = a * I(x, y) + b$$

The most common Pixel brightness transformation operations are

1. Gamma correction

It is a non-linear pixel adjustment and causes some adjustment on the image being altered.

$$S = c(r + \varepsilon)^\gamma$$

where ε is the offset, c and γ are positive constants

2. Sigmoid stretching

It is a continuous nonlinear activation function. By adjusting the contrast factor 'c' and threshold value, we may control the amount of illumination and darkening and hence the contrast enhancement of the entire image.

3. Histogram equalization

It is a contrast enhancement method and alters the intensity range so that histogram has the desired shape.

$$g(X) = af(X) + \beta$$

where gain and bias parameters

2) Geometrical Transformations

Geometric transformations change the positions of pixels but the colors should not vary [28]. The common operations are translation, rotation, scaling and distortion of images. Other special transformations are affine transformation and perspective transformations. Instead of taking shearing factors, scale factors and rotation angle, these transformations are merged into a single matrix and perspective transformation concentrates on the detailed information of a given image. In geometric transformation, the first step involved is spatial transformation of the physical arrangement of pixels and the second step is gray level interpolation which assigns black and white levels to the transformed image. These interpolation methods try to give a good estimation of the intensity of pixels and also are important in applying resizing/ resampling of image without pixel loss.

Some of the common methods for interpolation are Nearest neighbor interpolation, Bilinear interpolation and Bicubic interpolation. In the Nearest Neighbor method, the resultant pixel value is calculated by taking the nearest pixel value to the specified input. Bilinear interpolation method helps to estimate the values of the pixels in a suitable way while calculating the distance weighted average of the four nearest pixels which is placed diagonally. Bicubic interpolation finds the values of pixels by taking the average of 16 closest neighboring pixels.

3) Image Filtering and Segmentation

Filtering can be used to modify/enhance the properties and extract the needed information of the image like edges, corners and blobs. It will be defined by a kernel matrix which is an array and can be added with each pixel. Common filtering techniques are low pass filtering in which Smoothing is applied for reducing the noise by averaging nearby pixels. High pass filtering used to get the sharpening edges from an image. Directional filtering is a kind of edge detector which is used for first derivatives as slopes. It can be designed for all the directions. Laplacian filtering is used for second derivatives.

Segmentation is done for partitioning an image into individual regions depending on characteristics of the pixel values[11]. It is indirectly pointing to either separating the foreground from background image or grouping the regions based on color and shape of the pixel. Basically, there are two thresholding methods – Non-contextual thresholding and Contextual thresholding.

Non-contextual thresholding can convert grayscale to binary image[29]. Consider a binary region map with two regions, one with input values lesser than a threshold value and the other with input values greater than the threshold value.

$$T(x, y) = \mu(x, y) + K\sigma(x, y)$$

where μ -local mean, σ - standard deviation k- bias

1. Simple thresholding
2. Adaptive thresholding
3. Color thresholding

Contextual thresholding can be useful to separate individual objects by choosing the closeness of pixels. Discontinuity - based techniques can identify complete boundaries which are relative with uniform regions. Similarity based techniques can

be used to form uniform regions directly by grouping connected pixels.

There are 4 methods for Contextual segmentation.

1. Pixel connectivity
2. Region similarity
3. Region growing
4. Split-and-merge segmentation

Texture segmentation is the process of dividing an image into regions with different textures containing similar group of pixels. Some of the texture segmentation approaches are

1. Structural approach
2. Statistical approach
3. Model based approach
4. Filter based approach

B. Feature Extraction

It is used for the dimensionality reduction process. Raw data is divided into manageable groups[30]. It helps to find the best feature from data sets by selecting and joining variables into features. It is useful when it is needed to reduce the number of resources without destroying relevant information and also to reduce the amount of replicate data from the data set. It gives the relevant shape information which is present in a pattern[31]. It can help us to do classification processes efficiently. If its data sets are high dimensional, it has been converted into a representation of feature vector map. This phase can help in maximizing the recognition rate and in creating similar feature sets of the same symbol.

Some basic feature extraction methods use actual pixel values as feature[32-33]. In one such method, grey scale pixel values are combined with every pixel value in a sequence and a feature vector is generated. In another method, a new matrix is constructed with the mean value of pixels. In another method, edge features are used, where an edge is where there is a drastic change in pixel values, that is, the difference between pixel values is large in an edge.

The features thus extracted can be classified into local and global features[34]. Local features are generally geometric positions ie., number of endpoints, branches, joints, concave and convex parts. Global features are taken topological and statistical details ie. Number of holes, connectedness and invariant moments are all factors to consider. Finally, observe the characteristics: aspect ratio, number of endpoints, shape, size and number of loops, width and height distributions, slant, shape, average curvature, and gradient characteristics. Feature extraction techniques can be classified generally into three groups as follows:[35]

1. Statistical features: Zoning, Characteristic location, Crossing and distances
2. Global transformations and series expansion features: Fourier transform, Walsh hadamard transform, Rapid transform, Hough transform, Gabor transform, Wavelets, Karhunen-Loeve Expansion, Moments

3. Geometrical and topological features: Strokes, Stroke direction and Bays, Chain codes, End points.

Some of the commonly used feature extraction methods are tabulated with its description in Table 1. Some other feature extraction techniques are Template matching, Deformable templates, Unitary Image transforms, Graph description, Projection Histograms, Contour profiles, Zoning, Geometric moment invariants, Zernike Moments, Spline curve approximation, Fourier descriptors, Gradient feature and Gabor features.

TABLE I. : FEATURE EXTRACTION METHODS

S.No	Algorithm	Description
1	Principal Component Analysis	Unsupervised learning algorithm Linear dimensionality reduction Reduce the whole dataset into two dimensions.
2	Independent Component Analysis	Unsupervised learning algorithm Linear dimensionality reduction Reduce the whole dataset into three dimensions.
3	Linear Discriminant Analysis	Supervised learning Dimensionality reduction technique
4	Locally Linear Embedding	Nonlinear embedding High dimensional datasets
5	T-distribution stochastic Neighbor Embedding	Nonlinear embedding High dimensional datasets
6	Autoencoders	Non-linear transformations

C. Classification

This phase is a final stage of analyzing – predicting the specified classes[36-37]. The class is termed as target, label or categories. Supervised learning is applied when the recognition is based on the labeled class. In that, the system gets practiced with image by image and learns some strategy. According to the learned strategy, it starts doing the recognition. Unsupervised Learning is applied when the labels are not known. It tries to learn by its own by a grouping strategy. Based on this, it will recognize the character. Reinforcement Learning works by analyzing the past experience and makes own decision depending on the captured data. Moreover, it's a trial-and-error process.

1) Convolutional Neural Network (CNN)

A Neural Network is a collection of algorithms which are helpful to recognize pixel relationships in an input data. It should have an input layer, hidden layer and output layer. A neuron is a mathematical function, according to the network architecture it classifies the features and gives the best predictable results.

CNN is one of the best Deep neural networks to be used in image processing[38]. CNN is made up of more interconnected neurons that have weights and biases. It extracts higher level representations from the image. Preprocessing can be done by separate stages in neural networks. In CNN, it takes the raw pixel as input, analyzes the features on its own and creates a feature map ie. 3D matrix in which the size of the first 2D is the length and width of the images and the 3rd dimension is channels of a color image.

Layers in CNN:

1. Input Layer: It's like a buffer which holds the input and passes it to the corresponding layer[39-40].

2. Convolution Layer: In the convolution process, some filters can be applied to find new features ie. Output feature map

In that, performs multiplication and addition for all the elements to get a single value.

During the training period, it learns the optimal value for the filter matrix and tries to extract correct features based on texture, edges and shapes.

3. ReLu layer: Rectified Linear Unit is a kind of transformation to the output feature map. The output of the convolution layer is given to the ReLu layer where ReLu activation function is applied.

4 Pooling layer: It reduces the dimensions from the transformed feature map. It uses a sliding window and transforms it into the corresponding values, Min, Max, and Average pooling are used[41].

5. Fully connected layer: Here, every node is connected with every node in the next layer so that it can perform classification depending on the features extracted by the convolution.

These layers are shown in Fig. 4

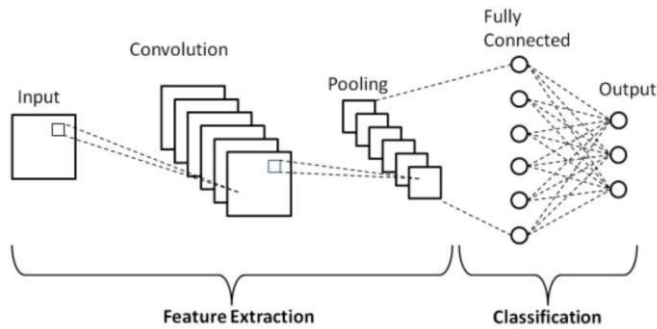


Fig. 4. Layers in CNN

Learning is done through supervised training [42]. The training is processed by two ways, i.e., forward pass and backward pass. In the forward path, error rate is calculated by comparing the network output with the training output. In the backward pass, the error is propagated backwards and weight factor and bias factor are adjusted to reduce the error.

IV. INFERENCES FROM THE LITERATURE REVIEW

There has been considerable research in applying deep learning methods for character recognition from stone inscriptions. The focus on such deep learning techniques has been driven by the perceived shortfall in the recognition rate of OCR techniques. This section presents the inferences from the literature review and outlines the issues and challenges which may help in the design of new mechanisms for character recognition.

A. Selecting a Suitable Preprocessing Approach

The aim of preprocessing is to enhance the contrast of the image, to normalize the brightness of the image, to suppress distortion and noise and to enhance some image features like edges which are needed for further stages[43-44]. Preprocessing does not increase the image information content. Instead, it suppresses information that is not relevant to the specific task and/or enhances image features that are relevant for further processing. It involves only pixel-wise operations and no high-level mechanisms are involved. From the wide range of techniques available for preprocessing, selecting the appropriate techniques for the image dataset is challenging.

B. Feature Extraction & Classification

Image features can refer to circles, lines, textures, shape of contours etc. [45]. The features to be extracted should be identified first and then the feature extraction network is to be designed suitably. CNN provides automatic feature extraction in the training phase. The feature extraction network comprises loads of convolutional layer and pooling layer pairs. Care should be taken to decide the number of feature extraction stages. The activation function to be used in the network should be chosen carefully [46].

There are more CNN architectures in the literature, like, AlexNet, LeNet, ResNet etc. Selecting the right architecture for the system under consideration is a challenge.

Deep architectures have millions of parameters which have to be learned during training[47]. The design of the deep CNN involves selection of number of neurons per layer, number of layers, neuron activation function and learning rate. The performance of the CNN varies with the depth of the network and with the addition of neurons in each layer. To extract meaningful features from raw data, huge amounts of training data is needed. If more training data is not available, then deep learning is not the appropriate technology [48].

C. Parameters for Performance Evaluation

Various parameters are used in the literature for analyzing the performance of automatic character recognition[49]. Some of the metrics are Mean Square Error (MSE), Peak Signal to Noise Ratio (PSNR), Maximum Difference (MD), Normalized Absolute Error (NAE), recognition rate (accuracy) and computational time[10]. A higher PSNR value indicates a higher quality reconstruction. Lower MSE and NAE values indicate a higher quality image. A greater MD score indicates that the image is of poor quality. Increasing the depth of the network, that is increasing the number of layers in the network will lead to improved recognition rate at the cost of increased computational time [50]. It is challenging to identify the best trade-off between recognition rate and computational time.

D. Scope of Recognition

From the literature review, it can be seen that some works have concentrated on recognizing vowels in Tamil, while some other works have focused on recognizing only consonants in Tamil. An efficient design which can recognize all the characters in Tamil is another challenge.

V. CONCLUSION & FUTURE WORK

A comprehensive study is presented to review the existing recognition methods for stone inscriptions on Tamil language. With the aforementioned review of literature, it is clear that CNN is more suitable for character recognition. Deep neural network is a fast-developing field of research and more new variants of CNN will quickly emerge in the foreseeable near future. Designing an appropriate deep neural network with the right set of parameters and with huge training dataset is the needed which will give higher recognition rate, lower error and will take lesser training time.

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S. Dhivya is a Research Scholar in the Department of Computing Technologies, SRM Institute of Science and Technology, Kattankulathur, India. She received her B.E. in Computer Science and Engineering from Adhiparasakthi Engineering College, Melmaruvathur, Anna University, India in 2012 and M.E. in Computer Science and Engineering from Adhiparasakthi Engineering College, Melmaruvathur, Anna University, India in 2014. She has 7 years of teaching. Her research interests include Machine Learning, Image processing and Internet of Things.



Dr. J. Rene Beulah is working as an Assistant Professor in the Department of Computing Technologies, SRM Institute of Science and Technology, Kattankulathur, India. She received her Bachelor of Engineering in Computer Science and Engineering from Manonmaniam Sundaranar University, Tirunelveli, India in 2004 and Master of Engineering in Computer Science and Engineering from Anna University, Chennai, India in 2006. She was awarded Ph.D. in the Faculty of Information and Communication Engineering from Anna University, Chennai in 2018. She has 9 years of teaching and 4 years of research experience. Her areas of research specialization are Machine Learning, Network Security and Data Mining.